

ENGINEERING PROJECT REVIEW

Smart night-time obstacle detection

A low-cost warning system for safer night driving.

Arduino UNO · Ultrasonic sensing · PIR movement detection
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01

PROBLEM STATEMENT

Night driving turns seconds into safety decisions

Detect early. Warn clearly.

When visibility drops, reaction time matters

- 1 Dim light reduces the driver's view, and a vehicle lamp or headlamp cutoff can create a sudden technical failure at the worst moment.
- 2 Animals and roadside obstacles can disappear into the darkness; delayed detection leaves less time to react and increases accident risk.
- 3 **Project objective:** detect hazards early and warn the driver in real time, even when visibility is reduced or lighting fails.

SYSTEM ARCHITECTURE

One low-cost system, four warning channels

Each layer has one job: sense the situation, decide its state, warn the driver, and coordinate the response.



Sense

Ultrasonic distance measurement and **PIR movement detection**.



Decide

Classify the situation as **safe**, **obstacle detected** or **critical**.



Warn

Red, yellow and green LEDs, buzzer and **16×2 I2C LCD**.



Control

Arduino UNO coordinates the response.

THE HARDWARE

The prototype combines familiar, accessible hardware

- 1 Arduino UNO
- 2 Ultrasonic sensor, PIR motion sensor and 16×2 I2C LCD
- 3 Red, yellow and green LEDs with resistors
- 4 Buzzer, breadboard, jumper wires and 5V power supply

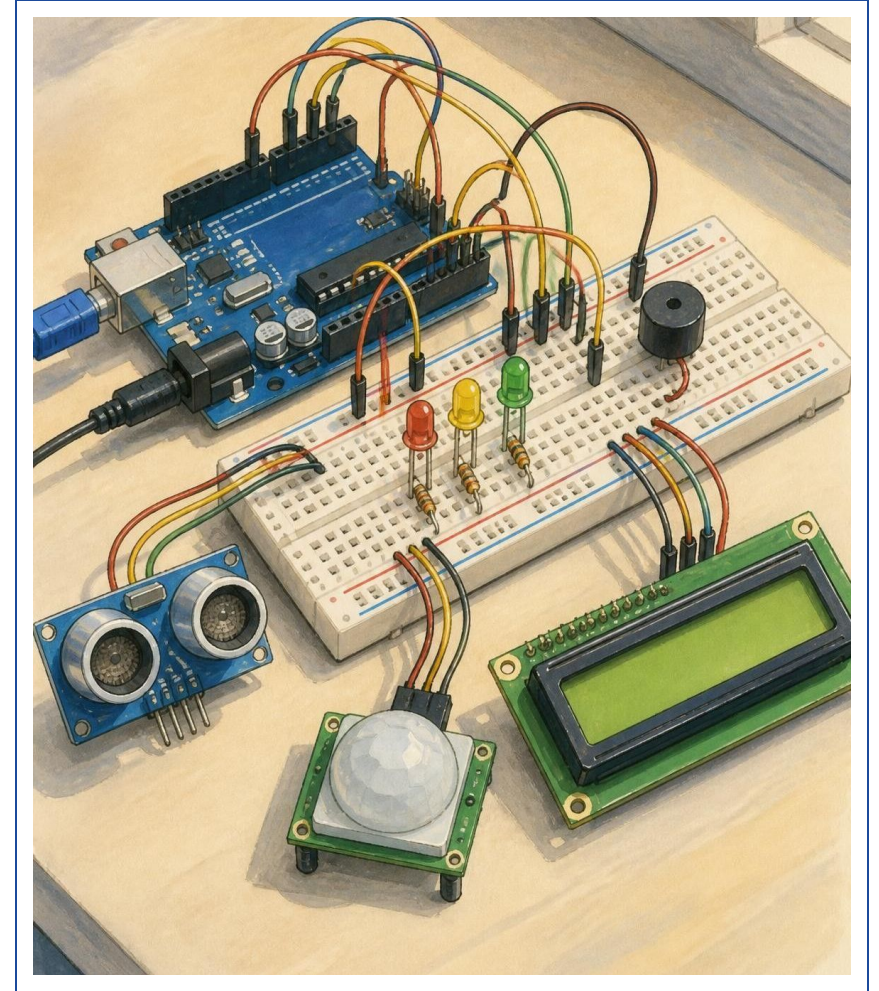


Fig. 4.1 Accessible hardware assembled on a breadboard

The sensing loop runs continuously

The Arduino repeats the same four actions so the warning state can change as the surroundings change.

01

Measure

Ultrasonic sensor measures the distance to the nearest obstacle.

02

Check

PIR sensor checks for movement in the monitored area.

03

Compare

Arduino compares the readings with warning thresholds.

04

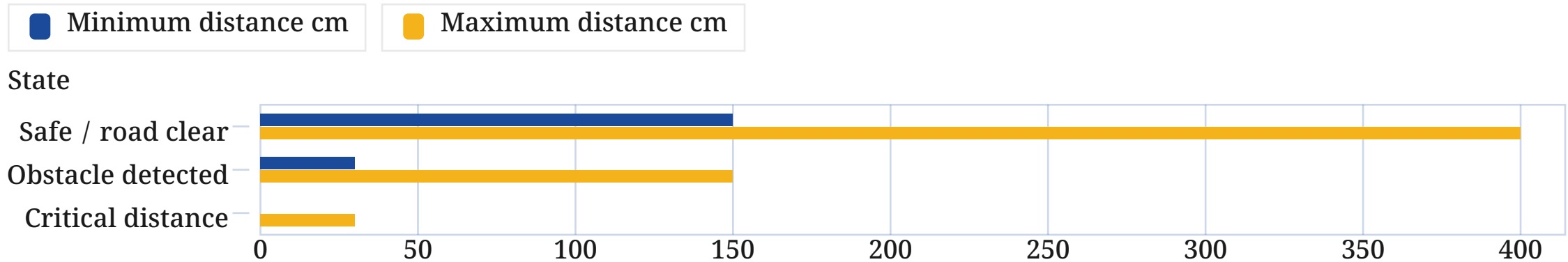
Update

LEDs, alarm and LCD update the driver immediately.

DISTANCE THRESHOLDS

Distance becomes a visible warning state

The system escalates from a calm green state to a critical red state as distance decreases.



Above 150 cm

Green LED, road clear.

30–150 cm

Yellow LED plus buzzer, obstacle detected.

Below 30 cm

Red LED plus high warning tone, critical distance.

PRIORITY ALERT

Movement overrides distance with an urgent alert

TRIGGER

PIR movement
detection

Triggers the red LED and high
warning tone.

DRIVER FEEDBACK

Distance + warning
status

The LCD displays the measured
distance and current warning
status.

SAME MESSAGE

Three feedback
paths

The driver receives the same
clear message through visual,
audible and text feedback.

CIRCUIT LOGIC

The circuit routes every signal back to the Arduino

Arduino UNO is the control point.

Signal	Connection
Power rails	Power rails distribute 5V and ground across the breadboard.
Digital lines	Digital lines connect the ultrasonic trigger/echo, PIR output, LEDs and buzzer.
I2C lines	I2C lines connect the LCD through SDA and SCL.
Current limiting	Each LED uses a resistor to limit current.

From local warnings to connected vehicle safety

A staged path from the working prototype to deeper vehicle integration — with each upgrade kept distinct from what is implemented today.

CURRENT Detect

Ultrasonic sensing identifies nearby obstacles; PIR sensing flags movement.

CURRENT Alert

LEDs, buzzer and LCD translate the detected state into a driver warning.

PLANNED Notify

Send warning status to a mobile endpoint through GPS/GSM connectivity.

PLANNED Acknowledge

Receive vehicle-side acknowledgement that the warning was received.

PLANNED Integrate

Explore ECU, braking or automatic power-control interfaces as later upgrades.

Scope boundary: the first two stages describe the bench-top prototype. Mobile/GPS-GSM notification, vehicle acknowledgement and ECU, braking or automatic power-control integration are planned extensions, not current performance claims.

FUTURE SCOPE

From warning device to automotive safety layer

Vehicle motor control

Control the vehicle motor when an object is detected.

User acknowledgement

Restore engine operation only after the user ends or acknowledges the incoming call.

Engine shutdown + notification

Turn off the engine and send a notification to the user.

V2X GPS/GSM location tracking

802.11p / C-V2X

Track the user's location through GPS with GSM or 4G and send an emergency notification.

TELEMETRY
4G / 5G / LTE

RADAR SENSOR

ULTRASONIC
SENSOR

ULTRASONIC
SENSOR

ULTRASONIC
SENSOR

CAN / LIN

MICROCONTROLLER